PlaceMux: 7-Day AI/ML Task Completion Report

This report breaks down exactly how we tackled our seven-day sprint to build the AI/ML intelligence layer for PlaceMux and how we transformed those daily tasks into a fully functional Streamlit website. Instead of just writing code in a vacuum, we took each day's deliverable and integrated it directly into a working web app.

Here is a look at what we accomplished each day:

Our Day-by-Day Progress

- **Day 1: Setting the Foundation.** We started by mapping out the core intelligence backbone, defining our skills ontology, the schema for our question bank, and how our APIs would communicate. This groundwork gave us the structure we needed for our item pools, testing models, and integrity reports.
- **Day 2: First Questions and Models.** We seeded our initial question bank (adding difficulty levels, Bloom's taxonomy tags, and item types) and chose the 3PL IRT model. We went with the 3PL model specifically because it accounts for the chance of students guessing correctly on multiple-choice questions.
- **Day 3: Building the Engine.** We brought our calibration to life by implementing the math that estimates a student's ability and updates their score after every response. This became the core Computer Adaptive Testing (CAT) engine that drives the live website.
- **Day 4: Assessment and Proctoring.** We refined the adaptive engine to always pick the most informative next question and know exactly when to stop the test. We also built the proctoring core, setting up identity verification, face detection, and risk flagging to monitor the session.
- **Day 5: Tying it Together.** We connected the assessment, scoring, and proctoring systems into one seamless flow. Instead of just automatically failing a student for a potential violation, we fused camera and behavioral signals into an integrity score that generates a review report for a fairer process.
- **Day 6: Meaningful Results.** We took those verified final scores and turned them into actionable advice. We mapped the students' demonstrated skills to specific job roles, ranking their best career matches and providing simple explanations for why those jobs were recommended.
- **Day 7: Testing and Polish.** We spent our final day running end-to-end dry runs to validate the reliability of our scoring, check our fairness assumptions, and ensure the proctoring behaved as expected. This resulted in a stable, demo-ready application.

How the Final Website Works

By the end of the sprint, we successfully built the final PlaceMux experience as a Streamlit application. It guides users through a complete, logical journey:

- **Home & Login:** The journey starts with a landing page explaining the platform, followed by a secure login that collects student details and initializes the proctored session.
- **The Assessment:** Students take an adaptive test driven by our CAT logic, which adjusts question difficulty on the fly and continuously updates their score until the test is complete.
- **Proctoring Dashboard:** Running in the background, this system checks identities, logs behavioral events, and monitors risk status.
- **Results & Recommendations:** Finally, the student receives their standardized score, performance band, and a detailed breakdown of their ranked career recommendations.

Conclusion

We successfully laid the AI/ML foundation for PlaceMux by tackling the seven assigned tasks step-by-step. Each day resulted in a functional piece of the intelligence layer, culminating in a cohesive website that handles adaptive testing, secure proctoring, fair scoring, and actionable career guidance all in one place.